Active Maintenance Report: coppock_kaur_2022

Coppock and Kaur (2022, American Journal of Political Science): Qualitative Imputation of Missing Potential Outcomes

2026-08-03

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This repository holds the actively maintained replication code for Coppock and Kaur (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1111/ajps.12697
Replication archive	10.7910/DVN/2IVKXD

The data are not redistributed here. The deposit is 10 files and about 0.7 MB, and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the exhaustive list of numbers the article and its supporting information print, the ground truth CSV is the comparison against the deposit and against the rewrite, and `build_ground_truth.R` is the gate that stops the run if a published number is checked by neither instrument. `coppock_kaur_2022_errata.pdf` at the root lists the numbers in the article that its own tables, appendix and data do not support. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_kaur_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 10 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, janitor, here, and for this report knitr and kableExtra. Paths resolve through `here`, so nothing depends on the working directory. Almost all of the run time is in the last two scripts, which simulate the bounds twenty-one times over in order to say what the deposited archive's method gives; the analysis itself is quick. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The figure PDFs always differ, because a PDF records the time it was written; the CSV and PNG output comes back byte-identical.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All five analysis scripts and the helpers file execute without error on R 4.6.0. Every package the archive names still installs from CRAN: tidyverse, xtable, janitor and reshape2. Nothing in the archive has been removed from the language; what has accumulated is deprecation. `geom_errorbarh()` and the `<ggplot> %+% data` idiom were deprecated in ggplot2 4.0.0, the `size` aesthetic for lines in 3.4.0, and `reshape2::melt` and `dcast` are superseded by tidyr. All of them still work and all of them warn. Everything that stands between this archive and a clean run is resolvable by substitution.

Running is not the same as reproducing, and one thing keeps the archive from reproducing its own figures exactly. `figures_2_3_A1_A2.R` estimates the bounds by simulation, 1,000 draws for Figures 2 and A1 and 10,000 for Figures 3 and A2, and it sets no seed. The published figures are labelled with rounded percentage points, so most of the simulation noise is absorbed by the rounding, but not all of it: repeating that simulation at twenty seeds moves at least one of the 70 labels the four figures determine off the value the bound actually takes in 14 of them. A reader running the deposited script today has better than even odds of seeing a figure that disagrees with the article somewhere, with no error and no warning. The maintained rewrite does not simulate these bounds at all, for the reason set out below.

Two smaller things. `table_4.R` prints its bounds as fractions where the paper prints percentage points, so the deposited output reads `[0, 0.3]` against the article's `[0, 30]`. It is a display inconsistency and not a numerical one. And the writing of every table and figure is commented out: each script builds the object and prints it, but the `xtable` and `ggsave` calls that would put it on disk are behind `#`. The archive is a set of scripts that show their work rather than a pipeline that produces the article's floats.

1.2 Does the maintained rewrite reproduce the paper?

Mostly, and the exceptions have one cause. 201 of the 227 verifiable ground truth claims match the published values to reported precision: every cell and bound of Table 3, every bound of Figure 1, every row of Table 4, the democratization row of Table 5, every label of Figures A1 and A2, all ten bounds of Figure 4, 761 of the 763 cells the two appendix case tables print, the abstract, and every in-text quantity from the Expert Survey section and Appendix A. The article's headline result is among them: the extreme value bounds on the ATE come to [-2, 49], 51 points wide, as published. A few of those matching labels are printed here at one decimal rather than as a whole number, because the bound falls exactly halfway between two whole numbers and no whole number states it; each agrees with the published label at the precision the page prints, and the table under *The unseeded simulation* lists every one.

The 26 that do not match fall into three groups.

17 of them follow from a single defect in the deposited data, described in full below: 7 imputation probabilities in `cases_clean.csv` do not match the probability the article's own appendix C states for that case, and the rewrite uses the stated values. Four are labels of Figure 2, twelve are labels of Figure 3, and one is the sentence summarising the ATT.

8 are places where the article contradicts itself. Seven of them are numbers in prose, spread over five sentences, two of which state a pair of numbers apiece; the rewrite agrees with the table or the appendix the sentence disagrees with in every case. The eighth is a cell of appendix Table B.2, an imputation probability the table prints differently from both appendix C and the deposited data.

The last is a second cell of appendix Table B.2, an observed outcome the deposited data record differently.

The remaining 5 recorded quantities have no comparison to make: four are the end-of-conflict row of the imputation summary, which the rewrite computes and neither the article nor the appendix prints, and one is a survey fact the deposit never recorded.

All of it is set out with corrected sentences and corrected floats in `coppock_kaur_2022_errata.pdf` at the root of this repository. **No conclusion of the article changes.** One published quantity moves materially: the average treatment effect on the treated, which the article states as a -15 percentage point effect and which is -22.5 points once the imputation probabilities appendix C states are the ones the analysis uses.

2 Paper overview

Citation: Coppock, A. and Kaur, D. (2022). "Qualitative imputation of missing potential outcomes." *American Journal of Political Science*, 66(3), 681-695. DOI: 10.1111/ajps.12697

Summary: The paper proposes a framework for meta-analysing qualitative causal inferences. A causal claim about a single case is a claim about the value of a missing potential outcome, so a set of such claims can be treated as imputations and fed into Manski (1999) extreme value bounds, which are the logical range of average causal effects consistent with what has been observed. Bounds on a binary outcome start 200 percentage points wide and narrow to 100 once the world reveals half the potential outcomes; each qualitative imputation narrows them further, and cases too murky to call are left unimputed rather than guessed. Where the analyst holds a probability rather than a point belief, the extension enumerates the 2^k consistent worlds and takes the probability-weighted bounds. The application is to 63 cases eligible for a transitional truth commission upon democratization, eight of which received one, with authoritarian resumption within ten years as the outcome. Imputation in four steps takes the bounds on the ATE to [-2, 49], 51 points wide. An email survey of country experts, 20 of whom responded, supplies an independent set of imputations; deferring to the experts where they answered narrows the bounds to 44 points. Appendix A repeats the exercise for 54 end-of-conflict cases.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
quimpo_helpers.R	Clean (sourced by the analysis scripts)	%>% to |>, map_df() to map() + bind_rows()
table_3_figure_1.R	Runs; deprecation warning	size= to linewidth=
table_4.R	Runs clean	Multiply the bounds by 100 to print percentage points, as the paper does
table_5.R	Runs clean	recode_factor() to case_when() + factor()
figures_2_3_A1_A2.R	Runs; three deprecation warnings; unseeded simulation	melt/dcast to pivot_longer/pivot_wider; geom_errorbarh() to geom_errorbar(orientation='y'); %+% to a named plot function; do() to reframe(); the bounds enumerated rather than simulated
figure_4.R	Runs; deprecation warning and a melt() id-variable warning	melt to pivot_longer with an explicit names_to; size= to linewidth=

3.1 The unseeded simulation, and the bounds it was estimating

The probabilistic extension is what makes Figures 2, 3, A1 and A2 stochastic in the archive. Where the authors hold a probability rather than a point belief about a missing potential outcome, the deposit's `sample_bounds()` draws binary potential outcomes from those probabilities and recomputes the bounds, and the reported estimate is the mean over draws. What the archive omits is `set.seed()`, so its figures are one unlabelled draw.

The maintained rewrite computes these bounds exactly, where the deposit estimates them by simulation. A bound is affine in the realised imputations: it is the count of realised ones among the imputed and observed entries, offset by the unimputed entries the bound fills with a constant, over the number of units. So the distribution the deposit samples from can be written down and enumerated instead, by convolving one Bernoulli imputation in at a time, and `bounds_distribution()` in `maintained/helpers.R` does that. The plotted estimate is the mean of that distribution and the interval its 2.5th and 97.5th quantiles.

This is not a different estimator. It is the same quantity by a different route, and it is the route the paper itself takes for the toy example, whose Table 4 enumerates all four consistent worlds and takes the probability-weighted bounds rather than sampling them. What it changes is threefold. Nothing in this repository depends on a seed, so the reproduction check is a byte comparison rather than a judgment about noise. The 84 bounds the four figures carry are now stated at their exact values rather than to within the simulation's error, which at the deposit's number of draws reaches 0.33 points at one of them. And a bound that falls exactly halfway between two whole numbers becomes visible as such, instead of being settled by whichever side a draw happened to land on.

`maintained/text_expected_bounds.R` keeps the comparison against the deposit's own method on the record: it runs that simulation once at a fixed seed, writes what it gives beside the exact value, and stops the build if any of the 84 bounds disagree by more than five standard errors of the simulated estimate. `maintained/text_seed_sensitivity.R` runs it at twenty seeds.

Table 2: The deposited archive's method, repeated at twenty seeds. Of the labels the four figures carry, the number the simulation rounds to a different whole number than the bound it is estimating does.

Labels rounded differently	Seeds
0	6
1	8
2	3
3	3

The size of the disturbance is one percentage point in the label, never more, and it barely touches the substantive claims: the final democratization bounds read [-2, 49] at 17 of the 20 seeds and [-1, 49] at the other 3, a difference of one point in the lower label. The point is not that the archive gets a different answer but that it gets a slightly different figure each time it is run, and it offers a reader no way to tell whether a discrepancy is noise or a mistake.

The table above counts only the 70 labels a whole number can state. The other 14 fall exactly halfway between two of them, and for those the deposit's method prints whichever side the draw fell on. The maintained figures print such a bound at the one decimal that states it.

Table 3: Bounds falling on a whole-number rounding boundary, with what one run of the deposit's method gives for the same quantity. All but the democratization ATT rows from Treated Cases onwards are a property of the deposit as published; those appear once the appendix C probabilities are restored.

Sample	Estimand	Step	Exact bound	Simulated, at seed 12345
Democratization	ATT	Initial Values	[-87.5, 12.5]	[-88, 12]
Democratization	ATT	Disbanded and Discredited Cases	[-87.5, 12.5]	[-88, 12]
Democratization	ATT	Treated Cases	[-22.5, -22.5]	[-22, -22]
Democratization	ATT	Non-transitional Cases	[-22.5, -22.5]	[-23, -23]
Democratization	ATT	Untreated Cases	[-22.5, -22.5]	[-22, -22]
Democratization	ATT	Unimputable Cases	[-22.5, -22.5]	[-22, -22]
End of conflict	ATU	Disbanded and Discredited Cases	[-27.5, 64.2]	[-27, 64]
End of conflict	ATU	Treated Cases	[-27.5, 64.2]	[-28, 64]

The 4 democratization ATT rows from *Treated Cases* onwards are the consequential ones, and they are the only rows in the table the corrections create. The ATT is exactly -22.5 once South Africa carries the probability appendix C states for it, so a whole-number label for it is not decided by the estimate at all: over twenty seeds the deposit's method prints [-22, -22] at 12 and [-23, -23] at 8. The rest of the table was on a boundary in the deposit as published, and the labels the article prints for them are one draw of a coin flip.

3.2 Checksums

Table 4: The deposit as Dataverse serves it with ?format=original. All ten files match the checksums Dataverse publishes, and all ten match the copy this program downloaded in March 2026.

File	Bytes	MD5 served (first 12)	Published MD5 agrees
cases_clean.csv	21433	0a238c073307	yes
codebook_cases_clean.pdf	348561	87154446febf	yes
figure_4.R	4282	d62dbb3dc25a	yes
figures_2_3_A1_A2.R	5859	b06b1c39bfba	yes
logfile.html	278521	831c13a4b09c	yes
quimpo_helpers.R	1517	876ec89e03d6	yes
README.txt	1712	634560b77b47	yes
table_3_figure_1.R	3456	e8fb68e7b478	yes
table_4.R	2220	237c27908efb	yes
table_5.R	1184	fd0efb3ad0e6	yes

Harvard Dataverse ingests `cases_clean.csv` as tabular data and derives a `.tab` representation from it, so the manifest records `originalFileName` and `originalFileSize` rather than the derived names and sizes, and `download_original.R` asks for `?format=original`. The published MD5 describes the deposited bytes, not the derived ones, and here it verifies them exactly. Other deposits in this program carry published checksums that verify neither file, so the download script checks against the served checksum and reports any published disagreement rather than failing on it.

4 Errata

Nothing in the deposited code is wrong. The deposited data are wrong in seven cells, and eight claims in the article's prose, spread over six sentences, are wrong on the article's own terms. All of it is set out with corrected sentences and corrected floats in `coppock_kaur_2022_errata.pdf` at the root of this repository. None of it changes a conclusion.

4.1 Seven imputation probabilities the deposit transcribes differently from appendix C

The imputed potential outcomes in this article are not measurements. They are qualitative judgments about what would have happened in each case, set out one by one in appendix C with the scholarship behind each of them, and the deposited `cases_clean.csv` is a transcription of those judgments. Appendix C states an imputation probability for 30 of the 31 imputed democratization cases; the exception, Thailand (1991-1992), refers to the entry above it rather than stating a number of its own. Twenty-three of the 30 agree with the deposit exactly. 7 do not.

Table 5: The imputation probabilities the deposited case file transcribes differently from the appendix C narrative that states them. The imputed value is the point imputation, which agrees with appendix C in every case.

Case	Appendix C entry	Imputed value	Probability in the deposit	Probability in appendix C
South Africa (1910-1994)	C.2.7	1	0.2	0.8
Ghana (1981-1993)	C.3.2	0	0.1	0.2
Uruguay (1973-1984)	C.3.6	1	1	0.9
Sierra Leone (1997-1998)	C.4.8	0	0.2	0.1
Nicaragua (1979-1990)	C.4.10	1	0.8	0.9
Central African Rep (1981-1993)	C.4.12	1	0.3	0.7
Sierra Leone (1992-1996)	C.4.13	1	0.3	0.7

3 of the 7 are exact complements, and in all three the deposit contradicts itself: an imputed potential outcome of 1 sits beside a probability below one half, which says the imputation is more likely to be 0 than 1. The main text states the South Africa probability too, on page 10, and states it as 0.8. The other 4 differ from the stated probability by one step of 0.1 and none of them crosses one half, so none changes an imputed value.

Appendix Table B.2 prints the deposited value in all 7 cases, so the published table inherits the transcription error rather than committing a second one. `maintained/table_b1_b2_case_dataset.R` is the one script in the rewrite that reads the deposit rather than the corrected file, precisely so that this stays checkable: the question those two tables answer is whether the published pages print what the deposit holds, and they do.

The rewrite uses the values appendix C states. `maintained/apply_appendix_c_corrections.R` is a visible, named step that writes both the corrected case file every analysis script reads and `output/appendix_c_corrections.csv`, the table above, so the deposited value sits beside the value that replaced it. It asserts the deposited value in every column it touches before overwriting it, so a change to the deposit stops the run rather than being absorbed. The probability columns are cumulative, and each correction is applied to every step column from the step at which the case is imputed onwards.

4.1.1 What it moves

The article's headline result is unchanged. The extreme value bounds on the ATE stay at [-2, 49] and 51 points wide, as do the abstract's three numbers, the expert survey comparison in Figure 4, Table 5, and every quantity in the end-of-conflict application, all seven corrections being democratization cases. The two intermediate steps of Figure 2 move by about one point in each label, and the ATU bounds of Figure 3 by about one and a half.

The consequential movement is in the ATT. Every treated case has both potential outcomes filled in by the second imputation step, which is why the ATT bounds collapse to a point, and South Africa is one of the eight treated cases. Restoring its probability from 0.2 to the 0.8 that both the main text and appendix C state moves the ATT from -15 to -22.5 percentage points.

Table 6: Every published bound the corrections move. Published values are the whole-number labels the figures carry; corrected values are exact. The third row is the article's headline result and rounds to the published label.

Quantity	Published	Corrected
Figure 2, Treated Cases	[-16, 68]	[-16.8, 67.3]
Figure 2, Non-transitional Cases	[-12, 62]	[-13.2, 61.4]
Figure 2, Untreated and Unimputable Cases	[-2, 49]	[-1.6, 49.2]
Figure 3, ATU, Untreated and Unimputable Cases	[0, 58]	[1.5, 59.6]
Figure 3, ATT, Treated Cases onwards	[-15, -15]	[-22.5, -22.5]

The article's ATU claim is untouched. It says the ATU bounds are "greater than 58 points wide", and that width is 58.2 points before the correction and after it, because both ATU bounds move by the same amount.

One imputation changes class. An imputation is made with certainty when its probability is exactly 0 or exactly 1, and Uruguay (1973-1984) is carried at 1 in the deposit where appendix

C entry C.3.6 states 0.9. The split the article states as "5 with certainty, 26 probabilistically" is 4 and 27 in the deposit, Uruguay being the difference, and 3 and 28 once appendix C is followed. The number left unimputed, 32, and the 31 imputed in total are the same either way.

4.1.2 One inconsistency this does not resolve

South Africa - Namibia (1966-1988), an end-of-conflict case, is the fourth place in the deposited data where a point imputation sits on the wrong side of its own probability: the imputed treated outcome is 0 and the probability that it equals 1 is 0.8. Appendix Table B.1 prints the same pair, so the table and the deposit agree. Appendix C covers the democratization cases only and Appendix A gives no case-by-case narrative for the end-of-conflict sample, so there is no stated probability anywhere to resolve it against, and the case is left exactly as deposited. The three democratization cases of the same shape are corrected above, because for those a stated probability exists.

4.2 Eight claims in the article's prose

The toy example's treated outcomes. The text introducing the toy example says "the outcome for three of the treated units and one of the untreated units is 1". Table 3 on the facing page shows units 1 through 4 treated with an observed outcome of 1, which is four, and the deposited `Y <- rep(c(1, 0, 1, 0), c(4, 3, 1, 2))` encodes four. Four is also what the bounds require: the initial bounds of [-40, 60] follow from four treated ones and would be [-50, 50] with three.

The probabilistic extension's probabilities. The text says that instead of imputing 0 for units 6 and 7, "we think the probabilities of being a '1' for units 6 and 7 are .3 and .4". Table 4 gives the four scenario probabilities as 0.56, 0.14, 0.24 and 0.06, which is the product distribution of 0.2 and 0.3, not of 0.3 and 0.4, which would give 0.42, 0.18, 0.28 and 0.12. The deposited `table_4.R` uses 0.2 and 0.3 and reproduces Table 4 exactly, as does the rewrite.

Appendix A's bounds width. Appendix A says the imputations shrink "the width of the extreme value bounds from 100 to 41 points" and then, in the next sentence, that "the bounds come to [-3, 42] (only 45 points wide)". Both cannot hold. The pipeline gives final end-of-conflict bounds of [-2.9, 41.5], a width of 44.4 points, which rounds to 44 and which is 45 if taken as the difference of the rounded endpoints, as the second sentence does. The 41 matches nothing.

How the 63 missing potential outcomes were imputed. The empirical section says "for all 63 unobserved potential outcomes, we imputed 5 with certainty, 26 probabilistically and we left 32 unimputed". Neither number holds under either reading of the data, as set out above. The 32 left unimputed and the 31 imputed in total are right.

The count of disbanded and discredited cases. Step 1 names two democratization cases, Bolivia and the Philippines, then says in the next sentence that "the observed outcome $Y_i(0)$ in each of these four cases was 0". Appendix Table B.2 lists two cases at that step. Appendix Table B.1 lists four end-of-conflict cases at the same step, which is where the count appears to have come from.

The ATT. "We can summarize the ATT as a -15 percentage point effect on return to authoritarianism." That is what the deposited data give; it is -22.5 under the appendix C probabilities.

4.3 Two appendix cells the deposit does not support

Appendix Tables B.1 and B.2 print the full case datasets, 763 cells between them, and the deposited case file reproduces 761.

Burundi's imputation probability. Table B.2 prints the imputed treated outcome for Burundi (1996-2005) as 1 with a probability of 0.9. Appendix C entry C.4.11 states 0.8 for the same case, and the deposited data carry 0.8. Deposit and appendix agree here, so there is nothing to correct in the data; the table cell is the outlier and no bound is affected.

Azerbaijan's observed outcome. Table B.2 prints an observed outcome of 0 for Azerbaijan (1991-1992), consistent with the untreated potential outcome of 0 printed beside it. The deposited `outcome` column records 1 for that case while its `y0_obs` column records 0, so the deposit contradicts the switching equation for this one case. Nothing in the analysis reads the `outcome` column, so nothing published depends on it, and the table matches what the analysis used.

5 Number-by-number comparison

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it.

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
table_3	column d	1,1,1,1,1,1,1,0,0,0	1,1,1,1,1,1,1,0,0,0	1,1,1,1,1,1,1,0,0,0	1	1
table_3	column Y	1,1,1,1,0,0,0,1,0,0	1,1,1,1,0,0,0,1,0,0	1,1,1,1,0,0,0,1,0,0	1	1
table_3	column Y0_initial	?,?,?,?,?,?,?,1,0,0	?,?,?,?,?,?,?,1,0,0	?,?,?,?,?,?,?,1,0,0	1	1
table_3	column Y1_initial	1,1,1,1,0,0,0,?,?,?	1,1,1,1,0,0,0,?,?,?	1,1,1,1,0,0,0,?,?,?	1	1
table_3	column Y0_easy	1,1,?,?,0,?,?,1,0,0	1,1,?,?,0,?,?,1,0,0	1,1,?,?,0,?,?,1,0,0	1	1
table_3	column Y1_easy	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1	1
table_3	column Y0_nulls	1,1,?,?,0,0,0,1,0,0	1,1,?,?,0,0,0,1,0,0	1,1,?,?,0,0,0,1,0,0	1	1
table_3	column Y1_nulls	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1	1
table_3	column Y0_hard	1,1,0,0,0,0,0,1,0,0	1,1,0,0,0,0,0,1,0,0	1,1,0,0,0,0,0,1,0,0	1	1
table_3	column Y1_hard	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1,1,1,1,0,0,0,1,0,?	1	1
table_3	EV bounds low, Initial Values	-40	-40	-40	1	1
table_3	EV bounds high, Initial Values	60	60	60	1	1
table_3	EV bounds low, Easy Cases	-20	-20	-20	1	1
table_3	EV bounds high, Easy Cases	30	30	30	1	1

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
table_3	EV bounds low, Nulls for 6&7	0	0	0	1	1
table_3	EV bounds high, Nulls for 6&7	30	30	30	1	1
table_3	EV bounds low, Hard Cases	20	20	20	1	1
table_3	EV bounds high, Hard Cases	30	30	30	1	1
figure_1	low bound, Before Any Data	-100	-100	-100	1	1
figure_1	high bound, Before Any Data	100	100	100	1	1
figure_1	low bound, Initial Values	-40	-40	-40	1	1
figure_1	high bound, Initial Values	60	60	60	1	1
figure_1	low bound, Easy Cases	-20	-20	-20	1	1
figure_1	high bound, Easy Cases	30	30	30	1	1
figure_1	low bound, Nulls for 6&7	0	0	0	1	1
figure_1	high bound, Nulls for 6&7	30	30	30	1	1
figure_1	low bound, Hard Cases	20	20	20	1	1
figure_1	high bound, Hard Cases	30	30	30	1	1
table_4	EV bounds low, unit 6 = 0, unit 7 = 0	0	0	0	1	1
table_4	EV bounds high, unit 6 = 0, unit 7 = 0	30	30	30	1	1
table_4	probability, unit 6 = 0, unit 7 = 0	0.56	0.56	0.56	1	1
table_4	EV bounds low, unit 6 = 1, unit 7 = 0	-10	-10	-10	1	1
table_4	EV bounds high, unit 6 = 1, unit 7 = 0	20	20	20	1	1
table_4	probability, unit 6 = 1, unit 7 = 0	0.14	0.14	0.14	1	1
table_4	EV bounds low, unit 6 = 0, unit 7 = 1	-10	-10	-10	1	1
table_4	EV bounds high, unit 6 = 0, unit 7 = 1	20	20	20	1	1
table_4	probability, unit 6 = 0, unit 7 = 1	0.24	0.24	0.24	1	1
table_4	EV bounds low, unit 6 = 1, unit 7 = 1	-20	-20	-20	1	1
table_4	EV bounds high, unit 6 = 1, unit 7 = 1	10	10	10	1	1
table_4	probability, unit 6 = 1, unit 7 = 1	0.06	0.06	0.06	1	1
table_4	probability-weighted point estimate, lower bound	-5		-5		1
table_4	probability-weighted point estimate, upper bound	25		25		1
table_4	2.5th quantile of the lower bound	-20		-20		1
table_4	97.5th quantile of the lower bound	0		0		1
table_4	2.5th quantile of the upper bound	10		10		1
table_4	97.5th quantile of the upper bound	30		30		1
table_4	probability that unit 6 has Y0 = 1	0.3	0.2	0.2	0	0
table_4	probability that unit 7 has Y0 = 1	0.4	0.3	0.3	0	0
table_5	democratization, Negative Effect	2% (1)	2% (1)	2% (1)	1	1
table_5	end of conflict, Negative Effect		0% (0)	0% (0)		
table_5	democratization, No Effect	40% (25)	40% (25)	40% (25)	1	1
table_5	end of conflict, No Effect		43% (23)	43% (23)		
table_5	democratization, Positive Effect	8% (5)	8% (5)	8% (5)	1	1
table_5	end of conflict, Positive Effect		13% (7)	13% (7)		
table_5	democratization, Unimputed	51% (32)	51% (32)	51% (32)	1	1
table_5	end of conflict, Unimputed		44% (24)	44% (24)		
figure_2	low bound, Before Any Data	-100	-100	-100	1	1
figure_2	high bound, Before Any Data	100	100	100	1	1
figure_2	low bound, Initial Values	-25	-25	-25	1	1
figure_2	high bound, Initial Values	75	75	75	1	1
figure_2	low bound, Disbanded and Discredited Cases	-25	-25	-25	1	1
figure_2	high bound, Disbanded and Discredited Cases	72	72	72	1	1
figure_2	low bound, Treated Cases	-16	-16	-17	1	0
figure_2	high bound, Treated Cases	68	68	67	1	0
figure_2	low bound, Non-transitional Cases	-12	-12	-13	1	0
figure_2	high bound, Non-transitional Cases	62	62	61	1	0
figure_2	low bound, Untreated Cases	-2	-2	-2	1	1
figure_2	high bound, Untreated Cases	49	49	49	1	1
figure_2	low bound, Unimputable Cases	-2	-2	-2	1	1
figure_2	high bound, Unimputable Cases	49	49	49	1	1
figure_a1	low bound, Before Any Data	-100	-100	-100	1	1
figure_a1	high bound, Before Any Data	100	100	100	1	1
figure_a1	low bound, Initial Values	-33	-33	-33	1	1
figure_a1	high bound, Initial Values	67	67	67	1	1
figure_a1	low bound, Disbanded and Discredited Cases	-32	-32	-32	1	1

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
figure_a1	high bound, Disbanded and Discredited Cases	61	61	61	1	1
figure_a1	low bound, Treated Cases	-26	-26	-26	1	1
figure_a1	high bound, Treated Cases	56	56	56	1	1
figure_a1	low bound, Non-transitional Cases	-24	-24	-24	1	1
figure_a1	high bound, Non-transitional Cases	54	54	54	1	1
figure_a1	low bound, Untreated Cases	-3	-3	-3	1	1
figure_a1	high bound, Untreated Cases	42	42	42	1	1
figure_a1	low bound, Unimputable Cases	-3	-3	-3	1	1
figure_a1	high bound, Unimputable Cases	42	42	42	1	1
figure_3	low bound, ATU, Before Any Data	-100	-100	-100	1	1
figure_3	high bound, ATU, Before Any Data	100	100	100	1	1
figure_3	low bound, ATU, Initial Values	-16	-16	-16	1	1
figure_3	high bound, ATU, Initial Values	84	84	84	1	1
figure_3	low bound, ATU, Disbanded and Discredited Cases	-16	-16	-16	1	1
figure_3	high bound, ATU, Disbanded and Discredited Cases	80	80	80	1	1
figure_3	low bound, ATU, Treated Cases	-16	-16	-16	1	1
figure_3	high bound, ATU, Treated Cases	80	80	80	1	1
figure_3	low bound, ATU, Non-transitional Cases	-12	-12	-12	1	1
figure_3	high bound, ATU, Non-transitional Cases	74	74	74	1	1
figure_3	low bound, ATU, Untreated Cases	0	0	1	1	0
figure_3	high bound, ATU, Untreated Cases	58	58	60	1	0
figure_3	low bound, ATU, Unimputable Cases	0	0	1	1	0
figure_3	high bound, ATU, Unimputable Cases	58	58	60	1	0
figure_3	low bound, ATT, Before Any Data	-100	-100	-100	1	1
figure_3	high bound, ATT, Before Any Data	100	100	100	1	1
figure_3	low bound, ATT, Initial Values	-88	-88	-87.5	1	1
figure_3	high bound, ATT, Initial Values	12	12	12.5	1	1
figure_3	low bound, ATT, Disbanded and Discredited Cases	-88	-88	-87.5	1	1
figure_3	high bound, ATT, Disbanded and Discredited Cases	12	12	12.5	1	1
figure_3	low bound, ATT, Treated Cases	-15	-15	-22.5	1	0
figure_3	high bound, ATT, Treated Cases	-15	-15	-22.5	1	0
figure_3	low bound, ATT, Non-transitional Cases	-15	-15	-22.5	1	0
figure_3	high bound, ATT, Non-transitional Cases	-15	-15	-22.5	1	0
figure_3	low bound, ATT, Untreated Cases	-15	-15	-22.5	1	0
figure_3	high bound, ATT, Untreated Cases	-15	-15	-22.5	1	0
figure_3	low bound, ATT, Unimputable Cases	-15	-15	-22.5	1	0
figure_3	high bound, ATT, Unimputable Cases	-15	-15	-22.5	1	0
figure_a2	low bound, ATU, Before Any Data	-100	-100	-100	1	1
figure_a2	high bound, ATU, Before Any Data	100	100	100	1	1
figure_a2	low bound, ATU, Initial Values	-29	-29	-29	1	1
figure_a2	high bound, ATU, Initial Values	71	71	71	1	1
figure_a2	low bound, ATU, Disbanded and Discredited Cases	-27	-27	-27.5	1	1
figure_a2	high bound, ATU, Disbanded and Discredited Cases	64	64	64	1	1
figure_a2	low bound, ATU, Treated Cases	-27	-27	-27.5	1	1
figure_a2	high bound, ATU, Treated Cases	64	64	64	1	1
figure_a2	low bound, ATU, Non-transitional Cases	-26	-26	-26	1	1
figure_a2	high bound, ATU, Non-transitional Cases	62	62	62	1	1
figure_a2	low bound, ATU, Untreated Cases	-2	-2	-2	1	1
figure_a2	high bound, ATU, Untreated Cases	48	48	48	1	1
figure_a2	low bound, ATU, Unimputable Cases	-2	-2	-2	1	1
figure_a2	high bound, ATU, Unimputable Cases	48	48	48	1	1
figure_a2	low bound, ATT, Before Any Data	-100	-100	-100	1	1
figure_a2	high bound, ATT, Before Any Data	100	100	100	1	1
figure_a2	low bound, ATT, Initial Values	-67	-67	-67	1	1
figure_a2	high bound, ATT, Initial Values	33	33	33	1	1
figure_a2	low bound, ATT, Disbanded and Discredited Cases	-67	-67	-67	1	1

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
figure_a2	high bound, ATT, Disbanded and Discredited Cases	33	33	33	1	1
figure_a2	low bound, ATT, Treated Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Treated Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Non-transitional Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Non-transitional Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Untreated Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Untreated Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Unimputable Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Unimputable Cases	-10	-10	-10	1	1
figure_4	low bound, us, The 20 Cases with Expert Responses	-10	-10	-10	1	1
figure_4	high bound, us, The 20 Cases with Expert Responses	40	40	40	1	1
figure_4	low bound, expert, The 20 Cases with Expert Responses	-15	-15	-15	1	1
figure_4	high bound, expert, The 20 Cases with Expert Responses	15	15	15	1	1
figure_4	low bound, us, All 63 Democratization Cases	-2	-2	-2	1	1
figure_4	high bound, us, All 63 Democratization Cases	49	49	49	1	1
figure_4	low bound, expert, All 63 Democratization Cases	-19	-19	-19	1	1
figure_4	high bound, expert, All 63 Democratization Cases	59	59	59	1	1
figure_4	low bound, combined, All 63 Democratization Cases	-3	-3	-3	1	1
figure_4	high bound, combined, All 63 Democratization Cases	41	41	41	1	1
abstract	bounds width prior to any analysis	100	100	100	1	1
abstract	bounds width after imputation	51	51	51	1	1
abstract	bounds width after the expert survey	44	44	44	1	1
text	democratization cases	63	63	63	1	1
text	treated democratization cases	8	8	8	1	1
text	bounds width before any data are collected	200	200	200	1	1
text	bounds width once half the potential outcomes are revealed	100	100	100	1	1
text	final ATE lower bound	-2	-2	-2	1	1
text	final ATE upper bound	49	49	49	1	1
text	final ATE bounds width	51	51	51	1	1
text	toy example: treated units with outcome 1	3	4	4	0	0
text	toy example: untreated units with outcome 1	1	1	1	1	1
text	toy example: bounds width before any data	200	200	200	1	1
text	toy example: final bounds width	10	10	10	1	1
text	ATU final bounds width	58	58	58	1	1
text	ATT summarised as a point effect	-15	-15	-22.5	1	0
text	cases imputed as a non-zero causal effect	6	6	6	1	1
text	expert responses received	20	20	20	1	1
text	democratization cases without an expert response	43	43	43	1	1
text	our bounds width, 20 responding cases	50	50	50	1	1
text	expert bounds width, 20 responding cases	30	30	30	1	1
text	our bounds width, all 63 cases	51	51	51	1	1
text	expert bounds width, all 63 cases	78	78	78	1	1
text	combined bounds width, all 63 cases	44	44	44	1	1
text	expert and author imputations agreed	7	7	7	1	1
text	expert and author imputations conflicted	3	3	3	1	1
text	we imputed, the experts did not	3	3	3	1	1
text	the experts imputed, we did not	7	7	7	1	1
text	cases the experts imputed	14	14	14	1	1
text	cases we imputed	10	10	10	1	1
text	experts who declined to impute	6	6	6	1	1
text	declining experts assigned to an untreated case	6	6	6	1	1
text	expert imputations differing from the observed outcome	0	0	0	1	1

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
appendix_a	end-of-conflict cases	54	54	54	1	1
appendix_a	treated end-of-conflict cases	6	6	6	1	1
appendix_a	bounds width once half the potential outcomes are revealed	100	100	100	1	1
appendix_a	bounds width after imputation	41	44	44	0	0
appendix_a	final ATE lower bound	-3	-3	-3	1	1
appendix_a	final ATE upper bound	42	42	42	1	1
appendix_a	final ATE bounds width from the rounded endpoints	45	45	45	1	1
appendix_a	ATT summarised as a point effect	-10	-10	-10	1	1
text	toy example: units	10		10		1
text	toy example: treated units	7		7		1
text	toy example: untreated units	3		3		1
text	toy example: easy imputations	5		5		1
text	toy example: untreated outcomes for units 1, 2 and 5	1, 1, 0		1, 1, 0		1
text	toy example: treated outcomes for units 8 and 9	1, 0		1, 0		1
text	probabilistic extension: possible sets of potential outcomes	4		4		1
text	unobserved potential outcomes	63		63		1
text	unobserved potential outcomes imputed with certainty	5		3		0
text	unobserved potential outcomes imputed probabilistically	26		28		0
text	unobserved potential outcomes left unimputed	32		32		1
text	disbanded and discredited democratization cases	4		2		0
text	observed outcome in the disbanded and discredited cases	0		0		1
text	South Africa: probability the imputed untreated outcome equals 1	0.8		0.8		1
text	disbanded cases: probability the imputed outcome equals 1	0.1		0.1		1
text	Nigeria: probability the imputed untreated outcome equals 1	0.1		0.1		1
text	experts disagreeing with our coding of treatment and outcome	5				
text	expert imputations conflicting with ours, restated in the discussion	3		3		1
text	end-of-conflict cases, stated in the main text footnote	54		54		1
text	unimputed cases as a share of the total			32 of 63, 50.8%		1
text	the bounds around the ATE include zero			[-2, 49] contains 0		1
text	the ATT bounds shrink to a point			[-22.5, -22.5], width 0		1
text	the ATU bounds are wider than the ATT bounds			58 against 0		1
text	the combined bounds are narrower than our original bounds			44 against 51		1
text	no expert assigned to a treated case declined to impute			0 of 6 declining experts		1
text	expert bounds are narrower than ours among the 20 responding cases			30 against 50		1
text	expert bounds are wider than ours among all 63 cases			78 against 51		1
table_b1	cells printed in the treatment indicator column	54		54		1
table_b1	cells printed in the observed outcome column	54		54		1
table_b1	cells printed in the revealed potential outcome column	54		54		1
table_b1	cells printed in the imputed untreated outcome column	54		54		1
table_b1	cells printed in the imputed treated outcome column	54		54		1
table_b1	cells printed in the imputation probability column	30		30		1
table_b1	cells printed in the unit-level effect column	54		54		1
table_b2	cells printed in the treatment indicator column	63		63		1

Table 7: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
table_b2	cells printed in the observed outcome column	63		62		0
table_b2	cells printed in the revealed potential outcome column	63		63		1
table_b2	cells printed in the imputed untreated outcome column	63		63		1
table_b2	cells printed in the imputed treated outcome column	63		63		1
table_b2	cells printed in the imputation probability column	31		30		0
table_b2	cells printed in the unit-level effect column	63		63		1

Of the 232 recorded claims, 181 can be compared against both the article and a deposited script, and 177 of those match. Six further claims are stated in the article but computed by no deposited script: the probability-weighted point estimate of the bounds in the probabilistic extension, [-5, 25], and the 2.5th and 97.5th quantiles of each bound, [-20, 0] and [10, 30]. The rewrite computes all six from the enumerated scenarios in `table_4_probabilistic.R` and all six match.

5.1 Coverage

The list of numbers to check is built from the article rather than from the pipeline, because a list built from the pipeline is silent exactly where the pipeline is missing something. `ground_truth/published_claims.csv` is every numeric token the article and its supporting information print, 268 of them, each classified by hand. 228 are quantities this pipeline can move, and each of those must be checked twice: once by a row in the ground truth and once by a block in `maintained/in_text_claims.R`, which reaches the same number by its own path through the same output files. The remaining 40 are scale endpoints, unit indices, period ranges and values copied out of other authors' papers, none of which any analysis can change.

Table 8: The published claims extraction, by class.

Class	Claims	Checked by
definitional	22	at the point of use
descriptive	9	ground truth and in_text_claims.R
pipeline	219	ground truth and in_text_claims.R
structural	14	at the point of use
transcribed	4	at the point of use

`ground_truth/build_ground_truth.R` enforces this. It sources the claims file rather than reading it as text, because a block that errors or prints nothing satisfies a textual check completely, and it counts the printed claims against the extraction. It also compares the two instruments value by value: they filter, convert and round independently, so a disagreement between them means one of the two is wrong. The run halts on any of these.

5.2 Appendix Tables B.1 and B.2

Appendix B prints the full case datasets, 54 end-of-conflict cases and 63 democratization cases, with seven columns each. The deposited archive has no script for either table, so until `table_b1_b2_case_dataset.R` was written the 763 published cells had nothing on this side to be compared against. All but 2 reproduce; both exceptions are described in the errata section above. That script is the one place in the rewrite that reads the deposit rather than the corrected case file, because the question it answers is whether the published pages print what the deposit holds. It also writes `table_b2_corrected_rows.csv`, the 7 rows of Table B.2 as published beside the same rows with the appendix C probabilities restored.

6 Maintained rewrite

The rewrite lives in `maintained/`: a helpers file, a corrections script, one cleaning script, four table scripts, four figure scripts and five in-text scripts. It is a translation rather than a reanalysis in every respect but two. `ev_bounds()`, `bounds_width()`, `sample_bounds()` and the probabilistic-extension functions are the paper's contribution and are carried over with their logic unchanged; what changes is the tidyverse around them. The first exception is `apply_appendix_c_corrections.R`, which restores the 7 imputation probabilities the deposit transcribes differently from the appendix that states them. The second is that the four bounds figures enumerate the distribution of each bound with `bounds_distribution()` rather than sampling it with `sample_bounds()`, which is the same quantity computed exactly instead of estimated. Both are described in the sections above.

6.1 Architecture

Table 9: The maintained rewrite.

Script	Output
helpers.R	none; packages, the QUIMPO functions and the toy example
apply_appendix_c_corrections.R	cases_corrected.rds, appendix_c_corrections.csv
clean_cases.R	cases_long.rds
table_3_toy_example.R	table_3_toy_example.csv, table_3_bounds.csv
table_4_probabilistic.R	table_4_probabilistic.csv, table_4_summary.csv
table_5_imputation_summary.R	table_5_imputation_summary.csv
table_b1_b2_case_dataset.R	table_b1_b2_case_dataset.csv, table_b2_corrected_rows.csv
figure_1_toy_bounds.R	figure_1_toy_bounds.pdf/.png/.csv
figures_2_a1_ate_bounds.R	figure_2_dem_ate_bounds.pdf/.png, figure_a1_eoc_ate_bounds.pdf/.png, figures_2_a1_gg_df.csv

figures_3_a2_att_atu_bounds.R	figure_3_dem_att_atu_bounds.pdf/.png, figure_a2_eoc_att_atu_bounds.pdf/.png, figures_3_a2_gg_df.csv
figure_4_expert_validation.R	figure_4_expert_validation.pdf/.png/.csv
text_expert_agreement.R	text_expert_agreement.csv, text_expert_agreement_counts.csv
text_imputation_counts.R	text_imputation_counts.csv
text_summary_stats.R	text_summary_stats.csv
text_expected_bounds.R	text_expected_bounds.csv
text_seed_sensitivity.R	text_seed_sensitivity.csv
in_text_claims.R	none; one printed line per published claim, checked by build_ground_truth.R

Three departures from a line-by-line translation are worth naming.

The toy example is defined once, in `quimpo_toy_example()` in `helpers.R`, rather than retyped in each of the three scripts that need it. In the archive, `table_3_figure_1.R` and `table_4.R` each set up the ten units and walk them through the imputation steps independently, which is the arrangement in which Table 3, Figure 1 and Table 4 can silently come to describe different examples.

Figures 2 and A1 come from one script and Figures 3 and A2 from another, because in each pair the two figures are two facets of one computation and splitting them would mean doing it twice. The archive's `figures_2_3_A1_A2.R` puts all four in one file; the rewrite splits along that boundary rather than the file boundary.

`text_summary_stats.R` reads the bounds back out of the figure output rather than recomputing them, so the widths quoted in the abstract and the body cannot drift from the figures they summarise. Every number it prints is traceable to a committed CSV, and no published value is typed into any script in `maintained/`.

6.2 Deprecated patterns replaced

Table 10: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
commented-out ‘setwd()‘	‘here::here()‘
magrittr pipe	native pipe
‘reshape2::melt()‘ / ‘dcast()‘	‘pivot_longer()‘ / ‘pivot_wider()‘
‘do(sample_bounds(...))‘	‘reframe(sample_bounds(...))‘
‘map_df(enframe, .id =)‘	‘map(enframe) |> bind_rows(.id =)‘
‘geom_errorbarh(height =)‘	‘geom_errorbar(orientation = ”y”, width =)‘
‘geom_linerange(size =)‘	‘geom_linerange(linewidth =)‘
‘<ggplot> %+% data‘	a named plot function taking the data as an argument
‘separate()‘	‘separate_wider_delim()‘
‘recode_factor()‘	‘case_when()‘ + ‘factor()‘

‘xtable‘ + ‘print.xtable‘ (commented out)	‘write_csv()‘ to ‘output/‘
‘ggsave()‘ (commented out)	‘ggsave()‘ to ‘output/‘, PDF and PNG
unseeded ‘rbinom()‘ simulation of the bounds	exact enumeration of the same distribution
bounds printed as fractions in ‘table_4.R‘	multiplied by 100, as the paper prints them

7 Figures

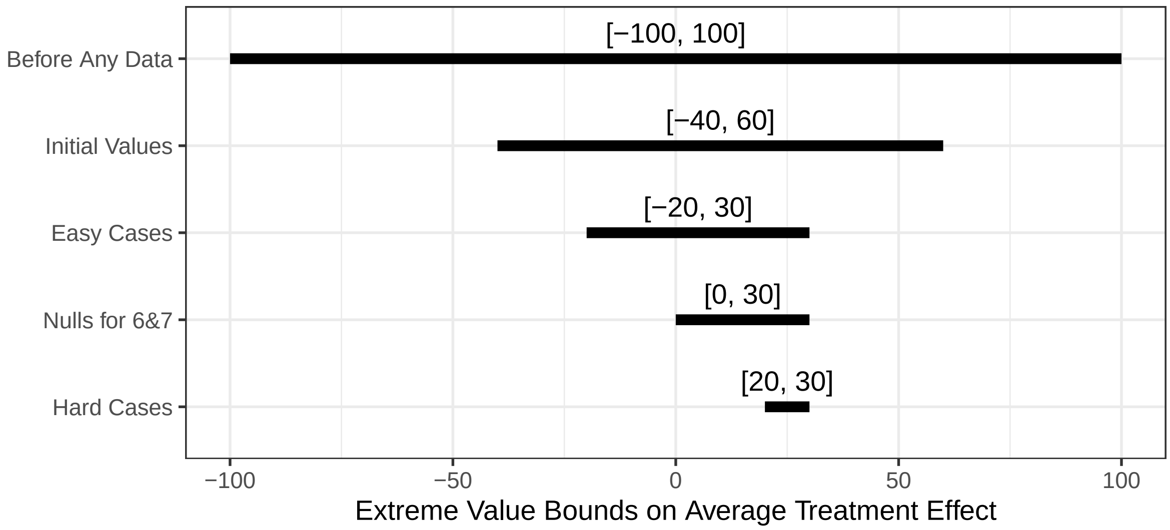

Figure 1: Figure 1: extreme value bounds at each step of the toy example.

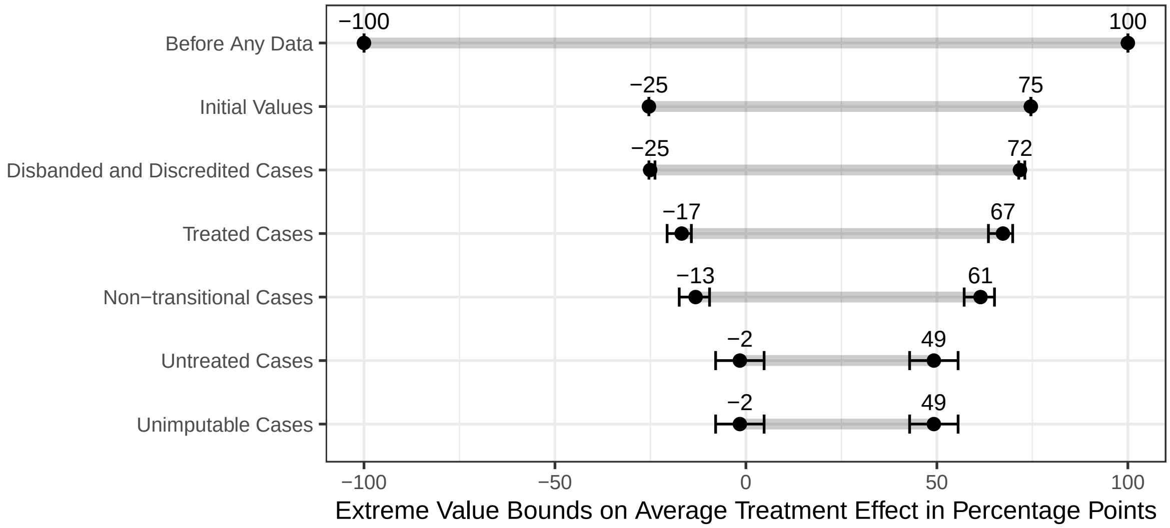

Figure 2: Figure 2: transitional truth commissions, extreme value bounds on the ATE.

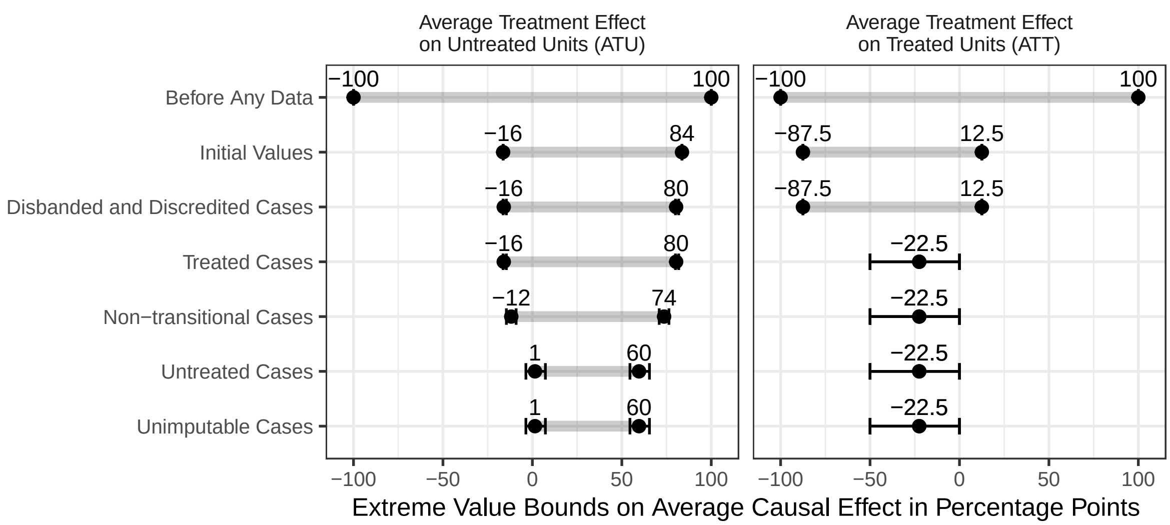

Figure 3: Figure 3: extreme value bounds by treatment status.

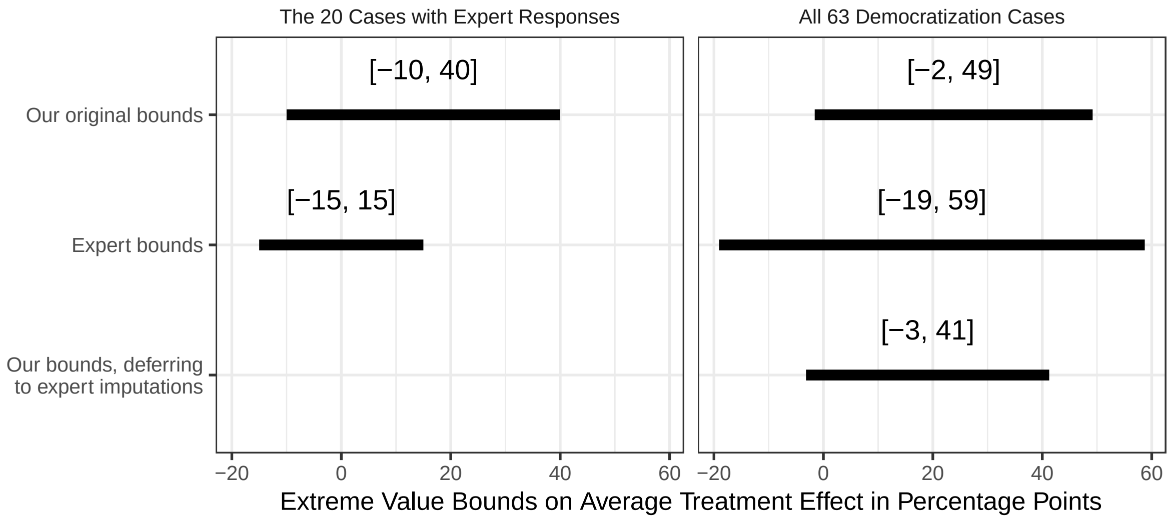

Figure 4: Figure 4: extreme value bounds implied by the expert imputations.

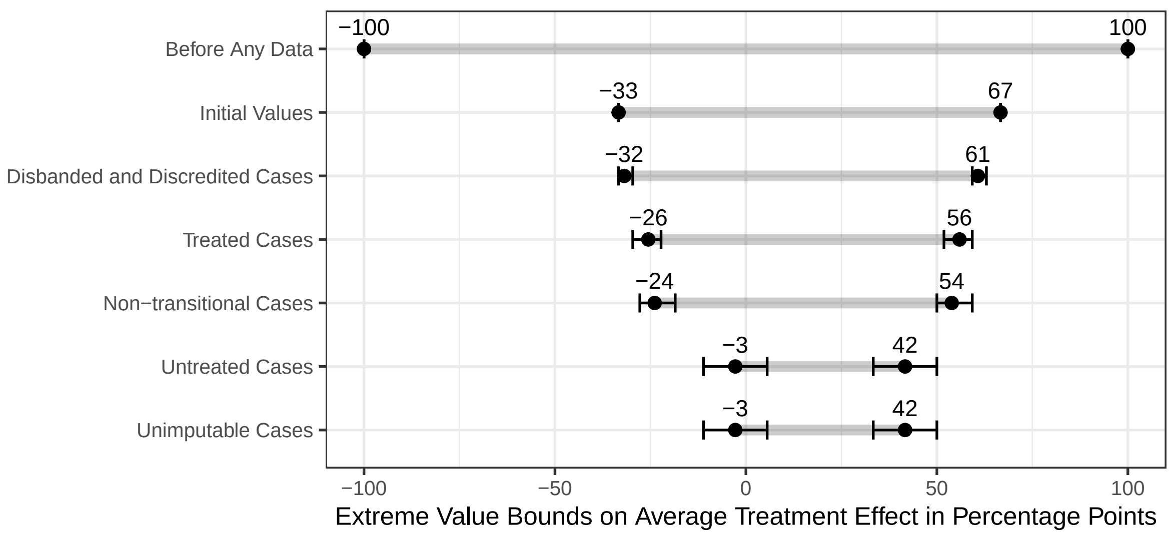

Figure 5: Figure A1: end-of-conflict cases, extreme value bounds on the ATE.

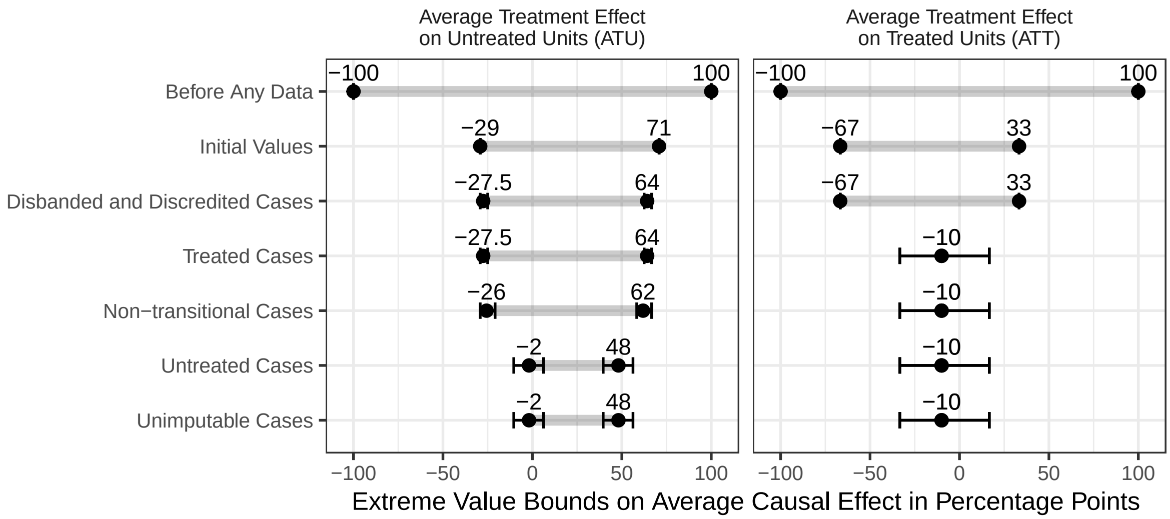

Figure 6: Figure A2: end-of-conflict cases, extreme value bounds by treatment status.

8 Expert survey verification

The Expert Survey section carries eleven quantities that no deposited script writes to disk, three of them recorded only in a code comment in `figure_4.R`. `text_expert_agreement.R` recomputes all of them from the deposited case data.

Table 11: The Expert Survey section, recomputed. Agreement means the two sides made the same imputation or both declined; the last row checks the claim that all 20 respondents either declined or imputed the observed outcome.

Quantity	Rewrite	Paper
agreement	7	7
disagreement	3	3
They declined, we imputed	3	3
They imputed, we declined	7	7
expert responses received	20	20
democratization cases without an expert response	43	43
cases the experts imputed	14	14
cases we imputed	10	10
experts who declined to impute	6	6
declining experts assigned to an untreated case	6	6
expert imputations differing from the observed outcome	0	0

The archive's `figure_4.R` computes the agreement coding and prints it with `table()`, then records the answer in three comment lines: "uruguay, burundi, south africa are 'disagreements' ", “7 cases, they imputed when we did not”, “3 cases, we imputed and they did not”. All three are correct. They are also the only place the numbers exist, which is the pattern this program treats as a defect regardless of whether the comment is right: a comment is not an output, and nothing checks it.

9 In-text quantities

Table 12: Bounds widths and summary effects quoted in the abstract, the body and Appendix A, read back out of the figure output that produced them.

Quantity	Rewrite	As a figure labels it
dem: bounds width before any data	200.00	200.0
dem: bounds width once the world reveals half the potential outcomes	100.00	100.0
dem: final ATE lower bound	-1.59	-2.0
dem: final ATE upper bound	49.21	49.0
dem: final ATE bounds width	50.79	51.0
dem: ATU final bounds width	58.18	58.0
dem: ATT final lower bound	-22.50	-22.5
dem: ATT final upper bound	-22.50	-22.5
expert: our bounds width, 20 responding cases	50.00	50.0
expert: expert bounds width, 20 responding cases	30.00	30.0
expert: our bounds width, all 63 cases	50.79	51.0
expert: expert bounds width, all 63 cases	77.78	78.0
expert: combined bounds width, all 63 cases	44.44	44.0
dem: cases imputed as a non-zero causal effect	6.00	6.0
eoc: bounds width once the world reveals half the potential outcomes	100.00	100.0
eoc: final ATE lower bound	-2.78	-3.0
eoc: final ATE upper bound	41.67	42.0
eoc: final ATE bounds width	44.44	44.0
eoc: ATT final lower bound	-10.00	-10.0
eoc: ATT final upper bound	-10.00	-10.0

The abstract and the body describe the same starting point in two ways, and both are right. The body says the bounds are 200 points wide “before any data are collected” and shrink to 100 “after the world reveals half the potential outcomes”. The abstract skips the first of those and says the bounds are 100 points wide “prior to any analysis”, meaning after data collection and before imputation. Both numbers are in the pipeline and both reproduce.

10 R environment

Table 13: Package versions used for this report.

Package	Version

tidyverse	2.0.0
here	1.0.2
janitor	2.2.1
knitr	1.51
kableExtra	1.4.0
ggplot2	4.0.3
dplyr	1.2.1
tidyr	1.3.2

R version: R version 4.6.0 (2026-04-24). The archive was written against R 4.1.2.